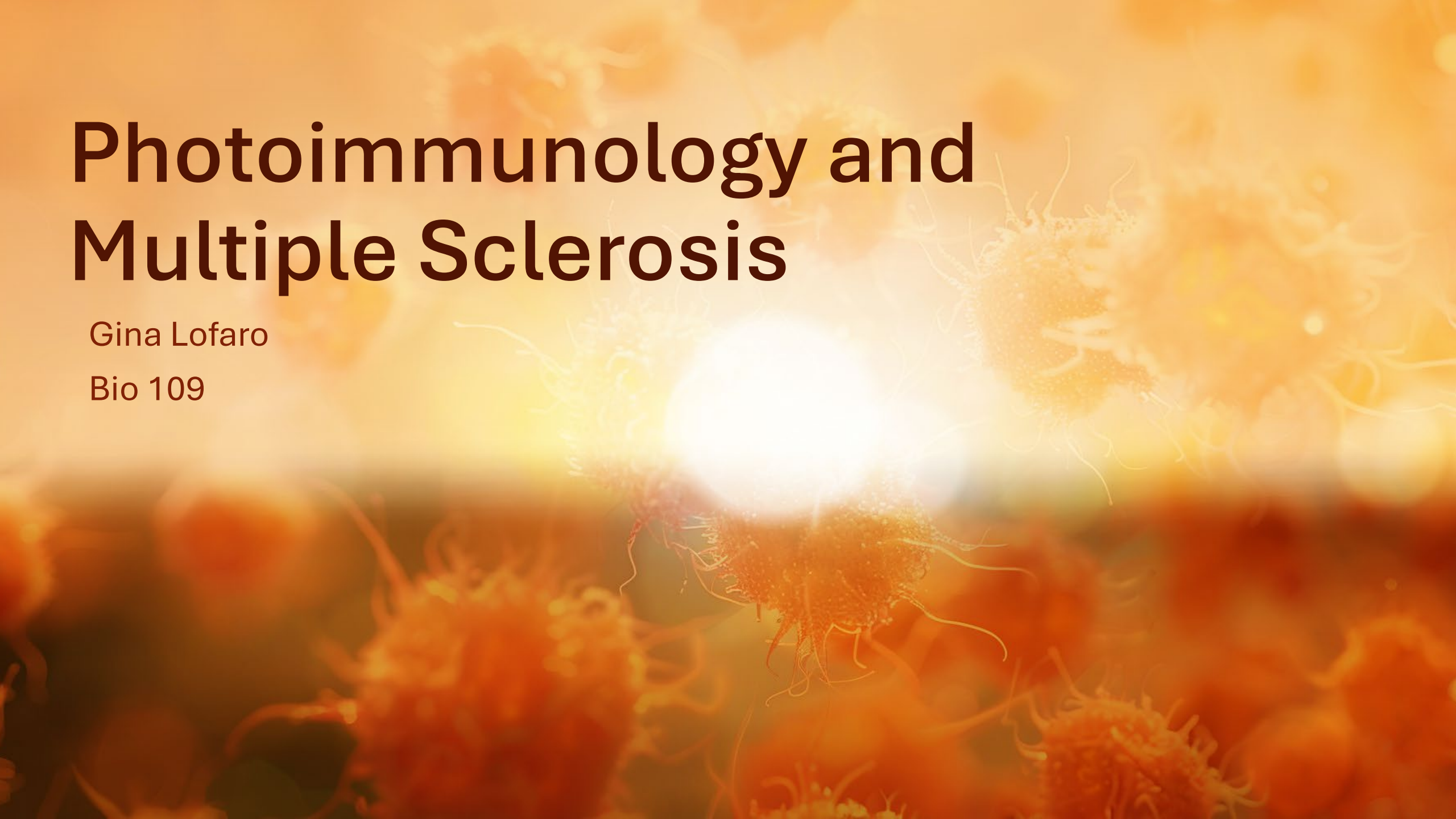


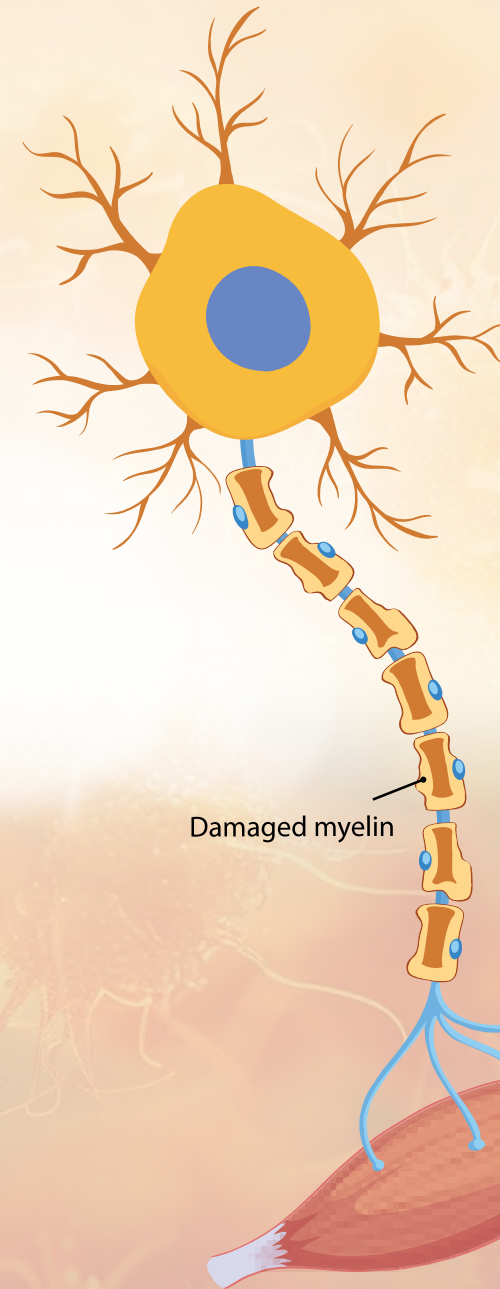
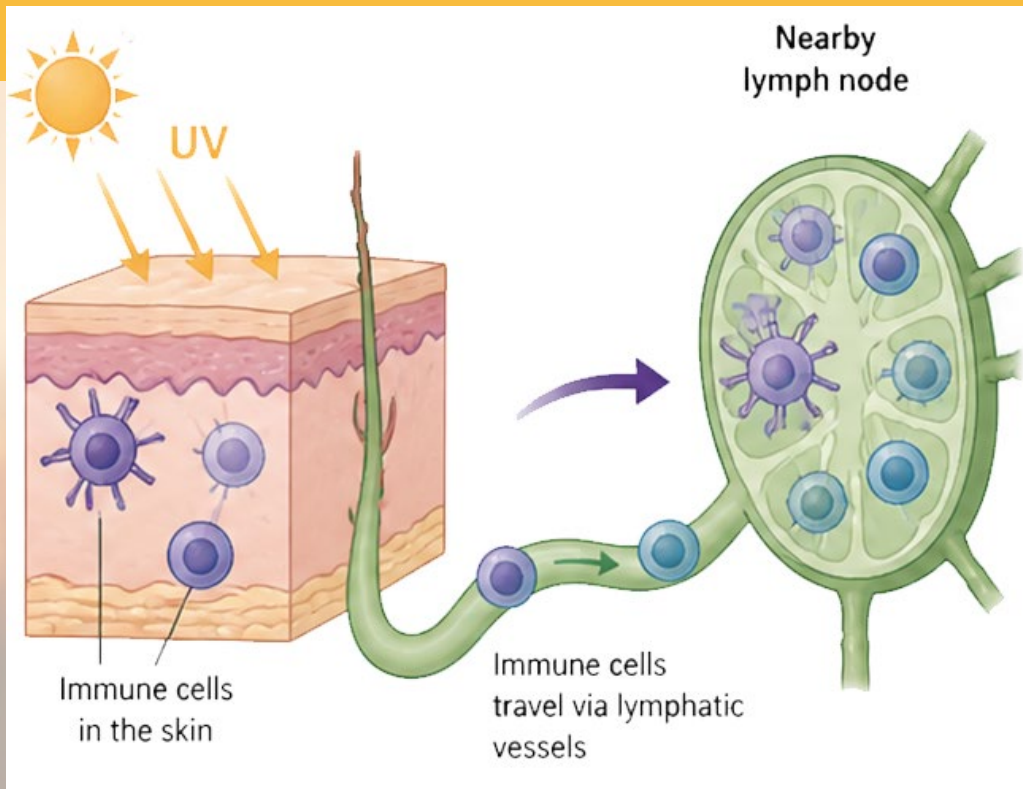
Photoimmunology and Multiple Sclerosis

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Bio 109



Photoimmunology is the scientific study of how ultraviolet (UV) radiation and visible light interact with the immune system

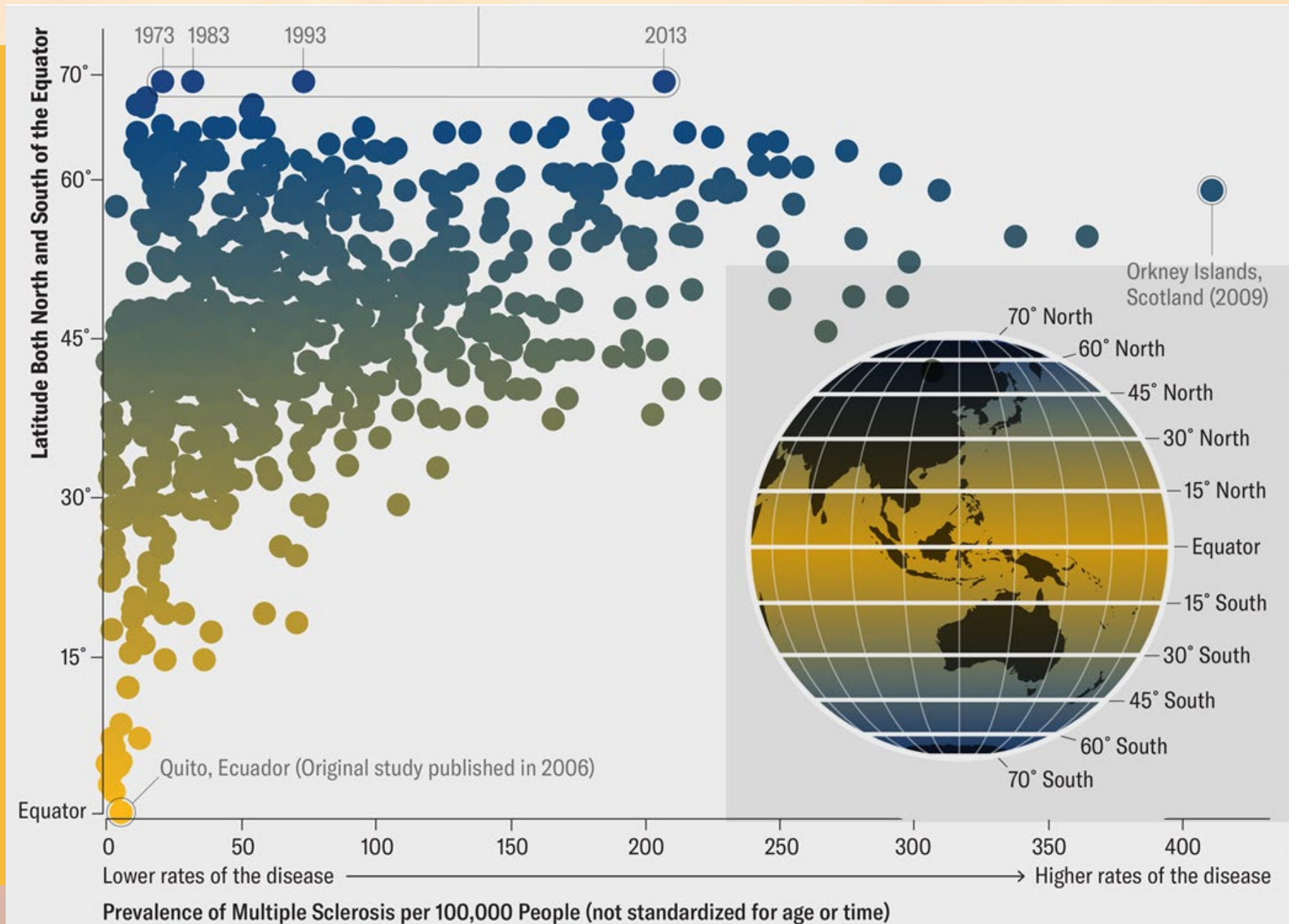


Multiple sclerosis (MS) is an autoimmune disease where the immune system attacks myelin, the insulating cover of nerve cells in the human body, causing damage to one's own central nervous system. It is a type of demyelinating disease, where the nervous system's ability to transmit signals is damaged. Symptoms can be physical, mental, or both.

The link between Photoimmunology and MS: latitude-gradient effect

MS is more prevalent in regions with lower natural UV exposure (farther from the equator)

Prevalence rates are close to zero near the equator and increase by 3.64 cases on average per 100,000 people for each degree of latitude, reaching well over 100 cases per 100,000 people in northern Europe and North America.

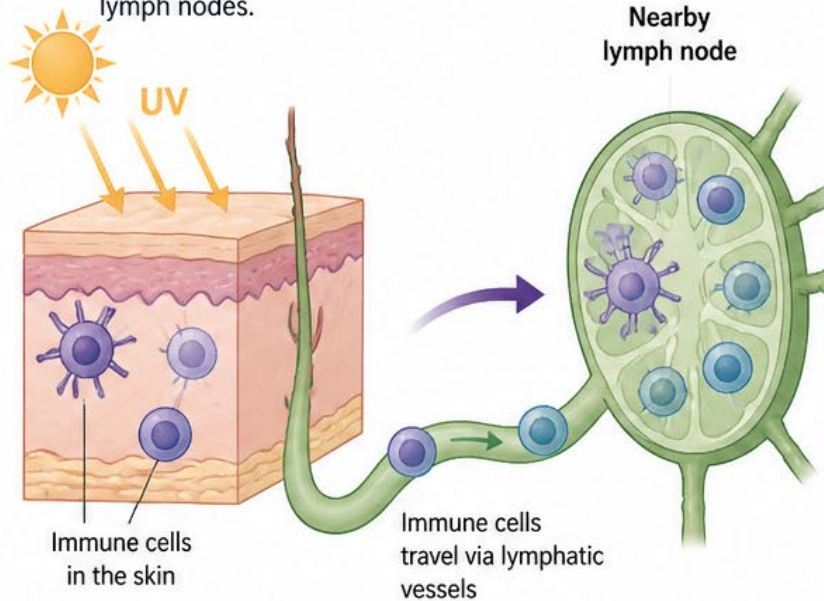


UV LIGHT, LYMPH NODES, AND MULTIPLE SCLEROSIS

UV exposure helps keep immune cells in lymph nodes, reducing inflammation and damage in MS.

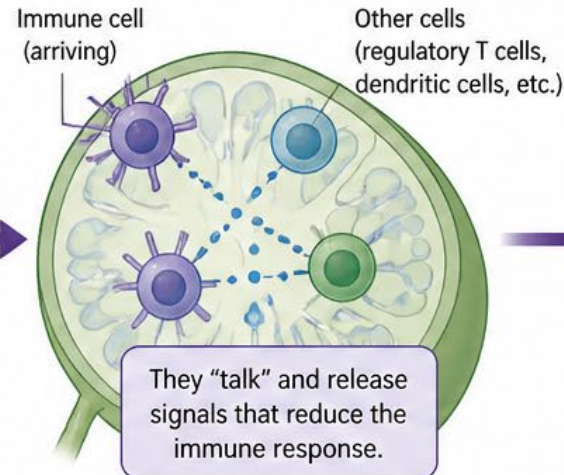
1 UV LIGHT HITS THE SKIN

When UV light hits the skin, immune cells in the skin travel to the nearby lymph nodes.



2 IN THE LYMPH NODE: IMMUNE CELLS ARE "TRAPPED" AND CALMED

In these "hubs" of immune activity, immune cells "talk" with many other cells to reduce the immune response.

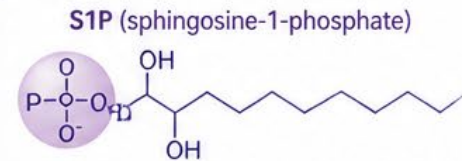


THE TRAPPING EFFECT

Trapping the immune cells means they cannot travel to other parts of the body to cause inflammation and damage, such as to the brain and spinal cord in MS.

3 THE LIPID SIGNAL BEHIND THE EFFECT

A fat (lipid) molecule called **sphingosine-1-phosphate (S1P)** was responsible for this trapping effect.



This provided the first evidence that UV can suppress the immune system by altering lipids in the lymph nodes.

BOTTOM LINE

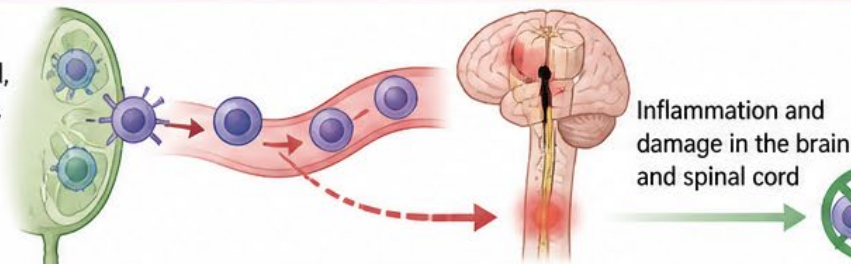
UV light → changes lipids in lymph nodes (↑ S1P)

↓
Immune cells are trapped & calmed

↓
Less inflammation and damage in the brain and spinal cord in Multiple Sclerosis

WITHOUT TRAPPING

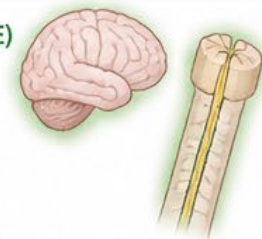
If immune cells are not trapped, they can leave the lymph node, travel in the bloodstream, and attack the brain and spinal cord in MS.



Inflammation and damage in the brain and spinal cord

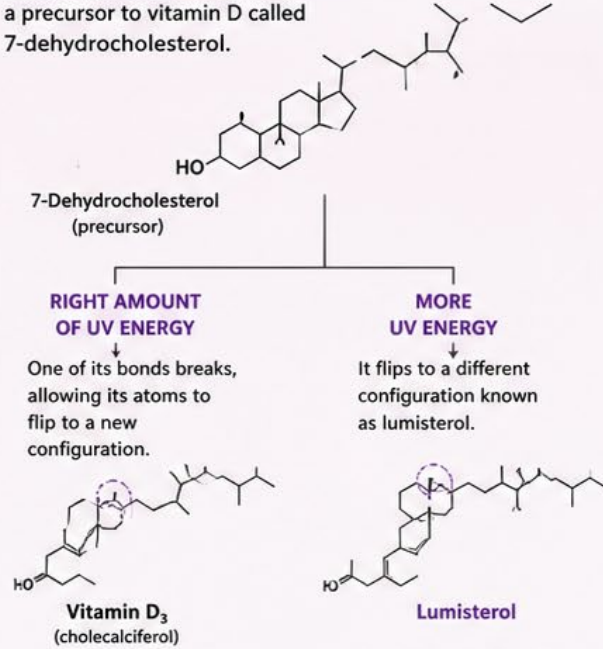
WITH TRAPPING (AFTER UV EXPOSURE)

Immune cells stay in the lymph node, so they cannot reach the brain or spinal cord. Less inflammation = less damage in MS.



1. 7-DEHYDROCHOLESTEROL PATHWAYS

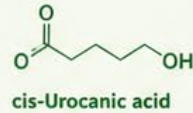
The body stocks the skin with a precursor to vitamin D called 7-dehydrocholesterol.



Lumisterol has been found in the blood in higher concentrations than vitamin D and has known anti-inflammatory and antitumor effects.

2. OTHER MOLECULES PRODUCED

The skin takes advantage of the bond-breaking power of UV radiation to produce multiple molecules.



Nitric oxide lowers blood pressure and reduces inflammation throughout the body.

3. PRO-OPIOMELANOCORTIN PATHWAY

Other skin cells respond to sunlight by increasing production of pro-opiomelanocortin (POMC).

PRO-OPIOMELANOCORTIN (POMC)

Cleaved by enzymes into three essential molecules:

β-ENDORPHIN



A neurotransmitter that causes feelings of well-being and reduces stress hormones.

ADRENOCORTICOTROPIC HORMONE (ACTH)



Triggers the release of cortisol, a steroid that regulates stress and suppresses inflammation.

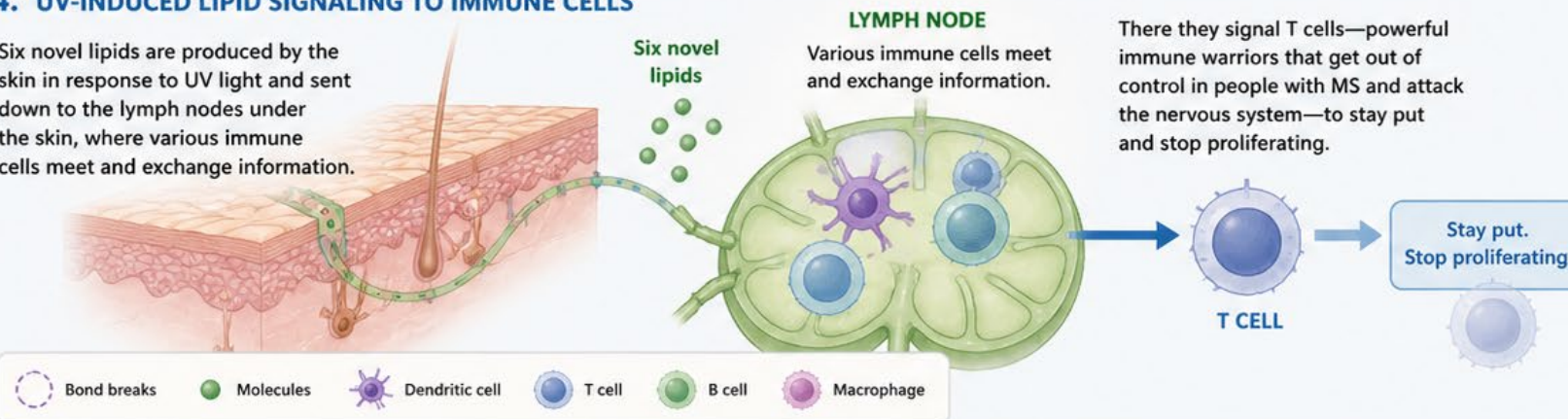
α-MELANOCYTE-STIMULATING HORMONE (α-MSH)



Repairs damaged cells, inhibits proinflammatory molecules and produces melanin to darken the skin.

4. UV-INDUCED LIPID SIGNALING TO IMMUNE CELLS

Six novel lipids are produced by the skin in response to UV light and sent down to the lymph nodes under the skin, where various immune cells meet and exchange information.



This pathway is separate from the one that suppresses cancer surveillance in the skin, which means it holds the promise of harnessing the good of UV radiation without the bad.